

## Multiple Choice (10 marks)

1. Most of the oxygen in the atmosphere is made by algae. Where do most of the algae live?
  - (a) coastal ocean
  - (b) deep ocean
  - (c) ponds
  - (d) lakes
2. In the design process, a trade-off occurs when a problem is solved but a feature is sacrificed. Which of the following is an example of a trade-off?
  - (a) A car's gas mileage is increased and the engine has less power.
  - (b) An airplane uses a more efficient engine and has higher performance.
  - (c) A sports drink's taste is improved and has the same nutritional content.
  - (d) A computer company upgrades the hardware and the price remains unchanged.
3. Lithospheric plates have moved to form land formations on the Earth's surface. Which land formation occurs as a result of the plates folding upon one another?
  - (a) an earthquake
  - (b) a mountain range
  - (c) a river
  - (d) a volcano
4. Which changes will most likely have a negative effect on an ecosystem?
  - (a) building a road
  - (b) planting a tree
  - (c) adding a freshwater source
  - (d) creating a nature sanctuary
5. Which geologic process most likely caused the formation of the Mount St. Helens Volcano?
  - (a) converging boundaries
  - (b) diverging boundaries
  - (c) transform faults
  - (d) rift zones

## True / False (10 marks)

*Answer True or False*

6. True or False: Designing a model is the most effective way to disprove a hypothesis about natural phenomena.
7. True or False: Aphids are first-level consumers in a food web because they feed on plants.
8. True or False: Light is refracted the least when passing through a tinted window compared to other optical devices.
9. True or False: Adding a base to a solution with a pH of 2 can increase its pH to above 7.
10. True or False: The sun would be enormous when compared to an average-sized star.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the forces acting on a fork that falls off a table, focusing on the primary force that causes it to move toward the floor. Explain your reasoning.
12. Discuss the importance of friction in bicycle ramps and analyze which materials can be used to enhance friction, providing examples and reasoning for your choices.

*End of Paper*